

API Endpoints for FarmVerse

Prepared by TeamA/Arindam

FARMS

1. ADD FARM

Endpoint: POST /farmverse/farms/addFarm

RequestBody:

```
{
  "farmName": "Name of farm",
  "farmType": "Type of farm",
  "areaSqMt": 5000.5,
  "soilType": "Loamy",
  "location": "Chennai"
}
```

ResponseBody:

SUCCESS

```
{
  "status": "ok",
  "statusCode": "200",
  "message": "Farm added Successfully",
  "id": "1"
}
```

FAILURE

```
{
  "status": "error",
  "statusCode": "400",
  "message": "Farm name is required",
  "id": null
}
```

2. UPDATE FARM

Endpoint: PUT /farmverse/farms/editFarm/{farmId}

RequestBody:

```
{
  "farmName": "New farm name",
}
```

```
"farmType": "New farm type",
"areaSqMt": 6000.0,
"soilType": "Clay",
"location": "Ranchi"
}
```

ResponseBody:

SUCCESS:

```
{
  "status": "ok",
  "statusCode": "200",
  "message": "Farm updated Successfully",
  "id": "5"
}
```

FAILURE:

```
{
  "status": "error",
  "statusCode": "404",
  "message": "Farm not found",
  "id": null
}
```

(Backend will check the farm's owner matches the logged-in farmer via JWT: a farmer shouldn't be able to edit someone else's farm by guessing an ID.)

3. DELETE FARM

Endpoint: DELETE /farmverse/farms/deleteFarm/{farmId}

RequestBody: NONE

ResponseBody:

SUCCESS

```
{
  "status": "ok",
  "statusCode": "200",
  "message": "Farm and its associated crops deleted successfully",
  "id": "5"
}
```

FAILURE

```
{
  "status": "error",
  "statusCode": "404",
  "message": "Farm not found",
  "id": null
}
```

Note:

This will be a Cascade delete: The crops if any within the farm to be deleted will ALSO be deleted.

Implementation note for later: this means the `crops` table's foreign key to `farm` needs `ON DELETE CASCADE` at the DB level (Postgres handles the cascade automatically)

4. VIEW FARM

Endpoint: `GET /farmverse/farms/viewFarm/{farmId}`

RequestBody: NONE

ResponseBody:

SUCCESS

```
{
  "status": "ok",
  "statusCode": "200",
  "message": "Farm details fetched successfully",
  "farm": {
    "farmId": 5,
    "farmName": "Green Valley Farm",
    "farmType": "Organic",
    "areaSqMt": 5000.5,
    "soilType": "Loamy",
    "location": "Ranchi",
    "crops": [
      { "cropId": 12, "cropName": "Wheat" },
      { "cropId": 13, "cropName": "Mustard" }
    ]
  }
}
```

FAILURE

```
{
  "status": "error",
  "statusCode": "404",
  "message": "Farm not found",
  "id": null
}
```

5. LIST ALL FARMS

Endpoint: GET /farmverse/farms/viewAllFarms

RequestBody: NONE (farmer identity from jwt)

ResponseBody:

```
{
  "status": "ok",
  "statusCode": "200",
  "message": "Farms fetched successfully",
  "farms": [
    {
      "farmId": 5,
      "farmName": "Green Valley Farm",
      "farmType": "Organic",
      "areaSqMt": 5000.5,
      "soilType": "Loamy",
      "location": "Ranchi",
      "cropCount": 2
    },
    {
      "farmId": 6,
      "farmName": "Sunrise Farm",
      "farmType": "Conventional",
      "areaSqMt": 3000.0,
      "soilType": "Sandy",
      "location": "Patna",
      "cropCount": 0
    }
  ]
}
```

Crops

1. ADD CROP

Endpoint: POST /farmverse/crops/addCrop

RequestBody:

```
{
  "farmId": 5, //To check the farm underwhich crop to be added
  "cropName": "Wheat",
  "cropType": "Rabi",
  "quantity": 500,
  "sowingDate": "20-01-2026",
  "harvestDate": "25-04-2026"
}
```

ResponseBody:

SUCCESS

```
{
  "status": "ok",
  "statusCode": "200",
  "message": "Crop added successfully",
  "id": "12"
}
```

FAILURE:

```
{
  "status": "error",
  "statusCode": "404",
  "message": "Farm not found",
  "id": null
}
```

2. EDIT CROP

Endpoint: PUT /farmverse/crops/editCrop/{cropId}

RequestBody:

```
{
  "cropName": "Wheat",
  "cropType": "Rabi",
  "quantity": 600,
  "sowingDate": "2026-01-10",
  "harvestDate": "2026-04-20"
}
// farm id not required since already in endpoint.
```

ResponseBody:

SUCCESS

```
{
  "status": "ok",
  "statusCode": "200",
  "message": "Crop updated successfully",
  "id": "12"
}
```

FAILURE

```
{
  "status": "error",
  "statusCode": "404",
  "message": "Crop not found",
}
```

```
"id": null
}
```

3. DELETE CROP

Endpoint: DELETE /farmverse/crops/deleteCrop/{cropId}

RequestBody: None

ResponseBody:

SUCCESS

```
{
  "status": "ok",
  "statusCode": "200",
  "message": "Crop deleted successfully",
  "id": "12"
}
```

FAILURE

```
{
  "status": "error",
  "statusCode": "404",
  "message": "Crop not found",
  "id": null
}
```

4. VIEW CROP

Endpoint: GET /farmverse/crops/viewCrop/{cropId}

RequestBody: NONE

ResponseBody:

SUCCESS

```
{
  "status": "ok",
  "statusCode": "200",
  "message": "Crop details fetched successfully",
  "crop": {
    "cropId": 12,
    "cropName": "Wheat",
    "cropType": "Rabi",
    "quantity": 500,
    "sowingDate": "2026-01-10",
    "harvestDate": "2026-04-15",
  }
}
```

```
    "farmName": "Green Valley Farm"
  }
}
```

FARMER

1. DASHBOARD

Endpoint: GET /farmverse/farmer/dashboard

Request: None (farmer identity from jwt)

ResponseBody:

SUCCESS

```
{
  "status": "ok",
  "statusCode": "200",
  "message": "Dashboard data fetched successfully",
  "fullName": "Full name here",
  "totalFarms": 2,
  "totalCrops": 5
}
```

ADMIN

DASHBOARD

Endpoint: GET /farmverse/admin/dashboard

RequestBody: NONE

ResponseBody:

SUCCESS

```
{
  "status": "ok",
  "statusCode": "200",
  "message": "Dashboard data fetched successfully",
  "totalFarmers": 24,
  "totalFarms": 61,
  "totalCrops": 143
}
```

1. EDIT FARMER

Endpoint: PUT /farmverse/admin/editFarmer/{username}

RequestBody:

```
{
  "fullName": "Ramesh Kumar Updated",
  "username": "ramesh_kumar",
  "email": "ramesh_new@example.com",
  "password": "NewSecurePass456"
}
```

ResponseBody:

SUCCESS

```
{
  "status": "ok",
  "statusCode": "200",
  "message": "Farmer updated successfully",
  "id": "8"
}
```

FAILURE (not found)

```
{
  "status": "error",
  "statusCode": "404",
  "message": "Farmer not found",
  "id": null
}
```

FAILURE (new username already taken)

```
{
  "status": "error",
  "statusCode": "400",
  "message": "Username already taken",
  "id": null
}
```

2. REMOVE FARMER

Endpoint: DELETE /farmverse/admin/removeFarmer/{username}

Request: NONE

ResponseBody:

SUCCESS (We'll be doing cascade delete: Farmer removed then all crops also)

```
{
  "status": "ok",
  "statusCode": "200",
  "message": "Farmer and all associated farms and crops removed successfully",
}
```



```
"id": "8"
}
```

FAILURE

```
{
  "status": "error",
  "statusCode": "404",
  "message": "Farmer not found",
  "id": null
}
```

3. ADD FARMER

Endpoint: POST /farmverse/admin/addFarmer

RequestBody:

```
{
  "fullName": "Ramesh Kumar",
  "username": "ramesh_k",
  "email": "ramesh@example.com",
  "password": "SecurePass123"
}
```

ResponseBody:

SUCCESS

```
{
  "status": "ok",
  "statusCode": "200",
  "message": "Farmer added successfully",
  "id": "8"
}
```

FAILURE (Username already taken)

```
{
  "status": "error",
  "statusCode": "400",
  "message": "Username already taken",
  "id": null
}
```

4. VIEW FARMERS

Endpoint: GET /farmverse/admin/viewFarmers

RequestBody: NONE

ResponseBody:

SUCCESS

```
{
  "status": "ok",
  "statusCode": "200",
  "message": "Farmers fetched successfully",
  "farmers": [
    {
      "fullName": "Arindam Pal",
      "username": "arindam123",
      "farmCount": 2
    },
    {
      "fullName": "Ramesh Kumar",
      "username": "ramesh_k",
      "farmCount": 5
    }
  ]
}
```